UNIT 3 · TOPIC 3(A)

# Testing of Fertilizer Applicators

*Syllabus topic: Testing of fertilizer applicators, sprayers and dusters*

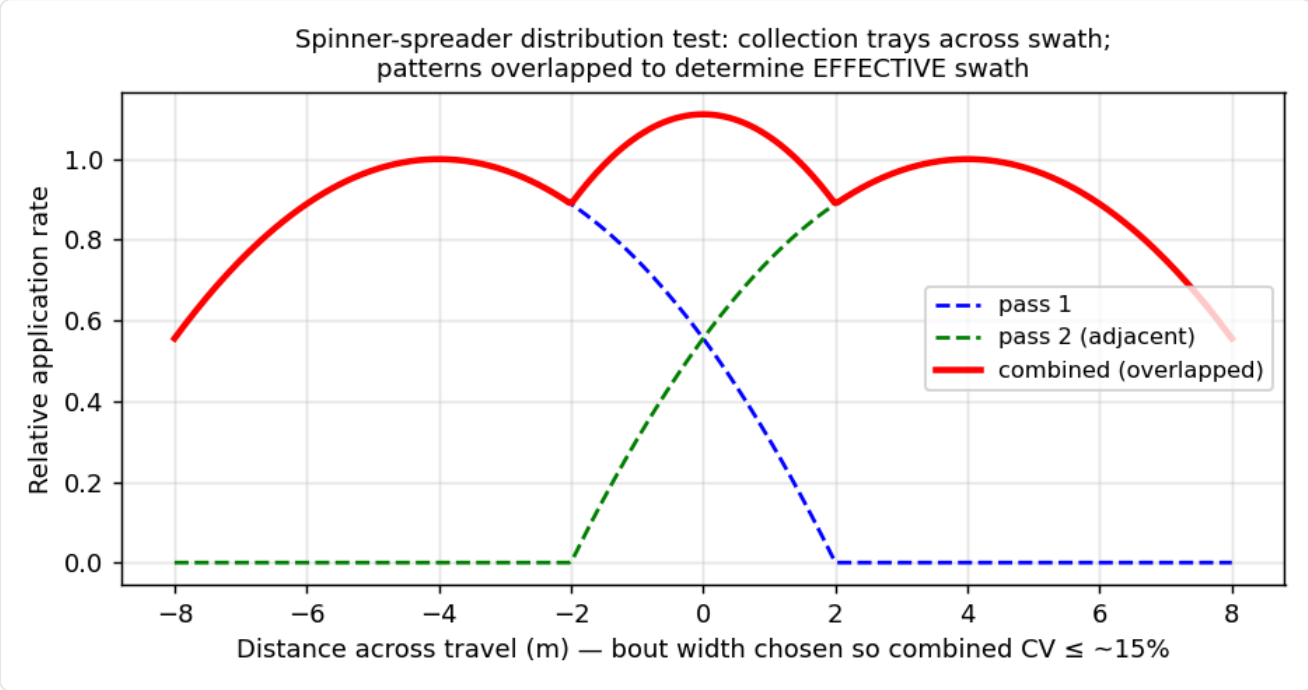
## Drill-type applicators

- Test the metering rate and per-row uniformity (as for drills) using the actual fertilizer.
- Check the effect of fill level, granule size and **moisture/hygroscopicity** (bridging, caking, corrosion of boxes).
- Check rate-lever repeatability.

## Broadcasters / spinner spreaders — distribution pattern test

1. Lay rows of collection trays across the swath.
2. Plot the single-pass pattern.
3. Overlap it mathematically for different bout widths.
4. Take the **effective swath** = the bout width at which the combined-pattern CV stays acceptable (≈ **≤15%**).

The pattern shape (triangular/oval/M-shaped) diagnoses spinner speed, drop point and vane wear.




*Spinner-spreader distribution test — patterns overlapped to find the effective swath*

## Practice MCQs

**Q55. Fertilizer-box materials are tested for:**

(a) Corrosion resistance
(b) Taste
(c) Transparency
(d) Magnetism

**Q56. The effective swath of a spinner spreader is fixed by:**

(a) Maximum throw distance

(b) Overlapped-pattern CV within acceptable limit (≈15%)

(c) Tank size

(d) PTO speed only

**Q57. Spreader distribution pattern is obtained using:**

(a) Collection trays across the swath

(b) Patternator grooves

(c) Penetrometer

(d) Belt

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**Answers**

Q55 — (a) Corrosion resistance

Q56 — (b) Overlapped-pattern CV within acceptable limit (≈15%) — *Single-pass spreader patterns are deliberately overlapped; the widest bout keeping combined CV ≤ ~15% is quoted as effective swath.*

Q57 — (a) Collection trays across the swath

**Sources:** RNAM (UN-ESCAP) Test Codes and Procedures for Farm Machinery; BIS Indian Standards for seed drills, sprayers and dusters; Kepner et al. (performance concepts); Jagdishwar Sahay; FMTTI (Budni) test procedures; NPTEL lectures.